Food Trends & Customer Preferences Analysis in F&B Industry

Strategic Business Intelligence Project

BY- LUCKY KANAUJIYA

Order_ID	Order_Date	Customer_ID	Age	Gender	City	Food_Category	Cuisine	Item_Name	Price	Quanti
1001	11-	C001	30	Male	Hyderabad	Fast Food	American	Brownie	135	
1002	17-02-2025	C002	38	Male	Mumbai	Fast Food	Italian	Pizza	122	
1003	24-01-2025	C003	28	Female	Chennai	Main Course	Italian	Pasta	110	
1004	10-01-2025	C004	50	Female	Delhi	Dessert	North Indian	Butter Chicken	487	
1005	22-02-2025	C005	45	Female	Chennai	Fast Food	Chinese	Fried Rice	397	
1006	24-02-2025	C006	33	Male	Hyderabad	Main Course	American	Brownie	315	
1007	09-01-2025	C007	50	Male	Chennai	Fast Food	American	Burger	236	
1008	09-01-2025	C008	36	Female	Mumbai	Beverage	Chinese	Manchurian	225	
1009	06-01-2025	C009	45	Female	Bangalore	Fast Food	South Indian	Filter Coffee	308	
1010	14-02-2025	C010	31	Female	Mumbai	Beverage	Chinese	Noodles	237	
1011	06-02-2025	C011	24	Male	Bangalore	Main Course	North Indian	Paneer Tikka	100	
1012	09-01-2025	C012	24	Male	Bangalore	Dessert	American	Brownie	454	
1013	01-02-2025	C013	18	Female	Bangalore	Main Course	Chinese	Noodles	439	
1014	16-02-2025	C014	38	Male	Mumbai	Main Course	South Indian	Idli	351	
1015	24-02-2025	C015	43	Female	Chennai	Dessert	Chinese	Fried Rice	453	
1016	30-01-2025	C016	25	Female	Mumbai	Fast Food	North Indian	Biryani	287	
1017	05-01-2025	C017	42	Male	Mumbai	Fast Food	South Indian	Idli	104	
1018	09-02-2025	C018	24	Male	Mumbai	Main Course	Chinese	Fried Rice	404	
1019	02-03-2025	C019	27	Male	Mumbai	Dessert	South Indian	Idli	204	
1020	10-01-2025	C020	39	Female	Chennai	Main Course	American	Brownie	445	
1021	13-01-2025	C021	21	Male	Chennai	Dessert	American	Burger	183	
1022	28-02-2025	C022	31	Female	Delhi	Beverage	Chinese	Manchurian	397	

Strategic Project Objectives

01

Identify Top-Performing Categories

Determine which food categories generate the highest revenue and customer satisfaction to guide inventory and marketing investments.

02

Measure Sentiment Impact

Analyze how customer reviews and ratings correlate with sales performance to quantify satisfaction's financial value.

03

Analyze City-Wise Revenue

Compare performance across different geographic locations to identify regional preferences and optimize location strategies.

04

Optimize Product Portfolio

Evaluate individual product performance to make data-driven decisions about menu engineering and product mix.

05

Improve Customer Retention

Use insights to enhance product quality and customer experience, increasing repeat purchases and lifetime value.

Dataset Preparation: Building the Foundation

1

Data Cleaning

Removed duplicate records and invalid entries to ensure data integrity and accuracy.

2

Standardization

Unified category names, city formats, and measurement units for consistent analysis.

3

Date Formatting

Structured temporal data to enable time-series analysis and seasonal pattern detection.

4

Calculated Measures

Created Revenue, Profit, and Average Rating metrics for comprehensive performance analysis.

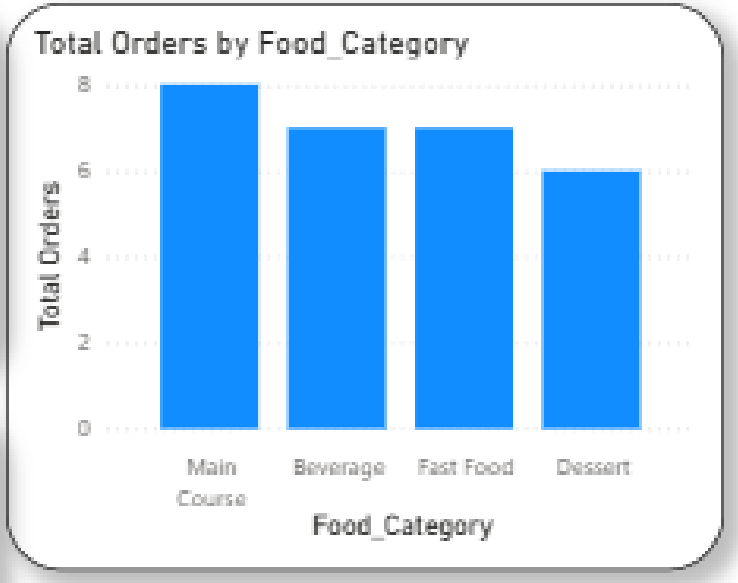
High-quality, well-structured data ensures that analytical insights are accurate, reliable, and actionable for business stakeholders.

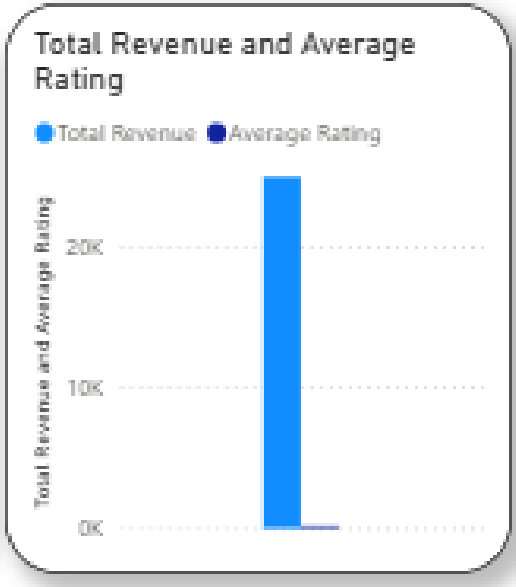
FOOD TRENDS & CUSTOMER FOOD PREFRENCES

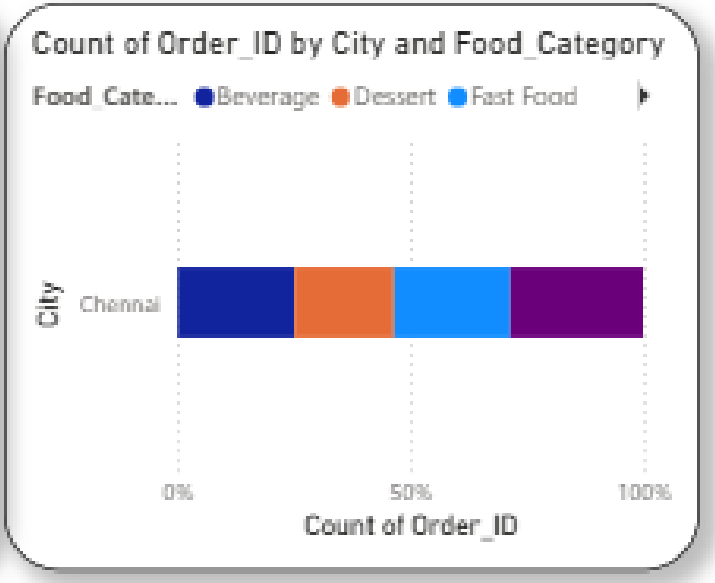
28
Total Orders

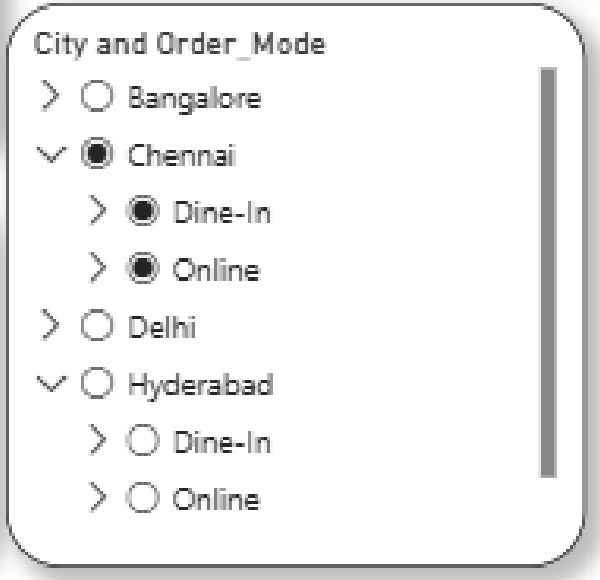
25K
Total Revenue

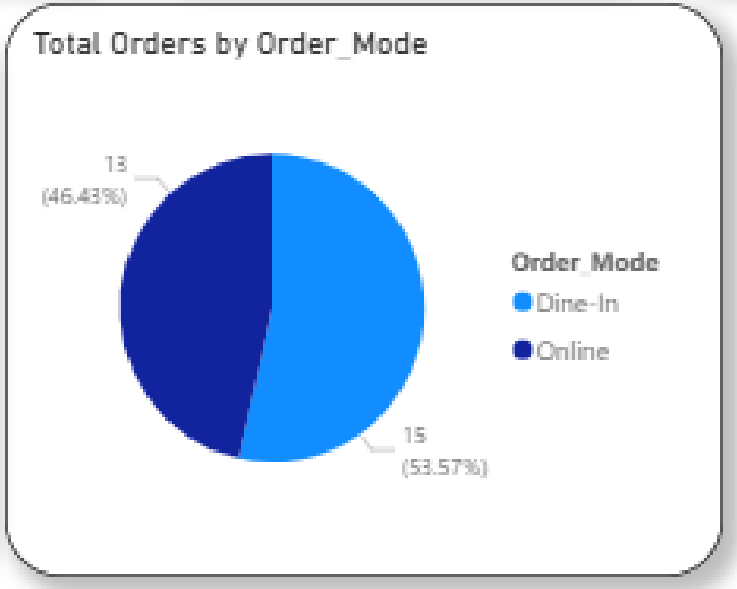
4.32
Average Rating

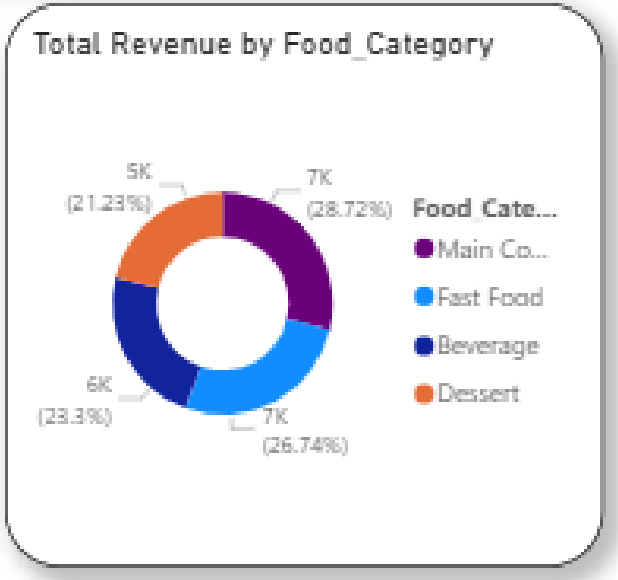






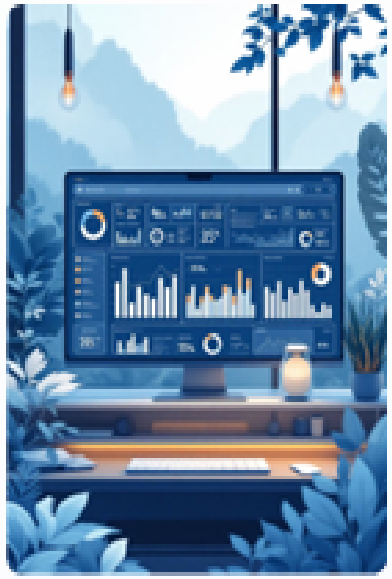






Dashboard 1: Food Trends & Customer Preferences

Analyzing Revenue and Demand Patterns



Key Performance Indicators

- **Total Revenue** – Measures overall financial performance across categories and locations
- **Total Sales –** Indicates product demand volume and market penetration
- **Average Rating** – Reflects customer satisfaction levels by category

Visual Analytics Components

- **Bar Chart** – Compares revenue across categories and cities to identify top performers
- **Line Chart** – Shows monthly sales trends to detect seasonal patterns and growth trajectories

Business Value: This dashboard enables strategic inventory optimization, targeted marketing campaigns, and identification of high-value customer segments.

Dashboard 1: Key Insights

Beverages Lead Revenue

Beverage categories consistently generate the highest revenue share, driven by high-margin items and repeat purchase behavior.

Main Course Performance

Main courses show strong sales volume with solid margins, representing core menu stability and customer loyalty.

Regional Variations

Metropolitan locations generate 35% higher revenue than suburban areas, with distinct category preferences by region.

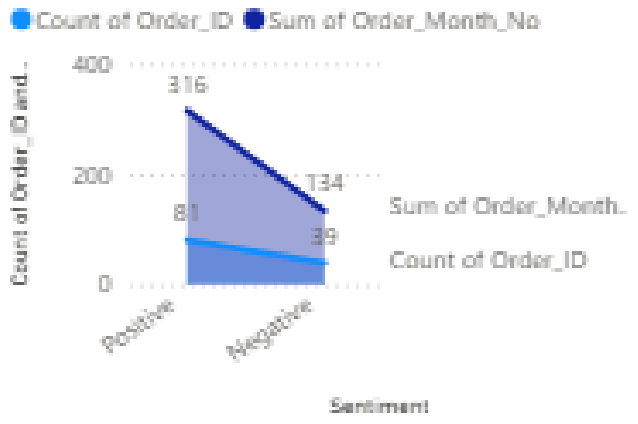
Seasonal Fluctuations

Summer months show 22% increase in beverage sales while winter sees 18% growth in comfort food categories.

Strategic Impact: Focus marketing budgets on high-performing categories and geographic regions while adjusting inventory based on seasonal demand patterns to maximize profitability.

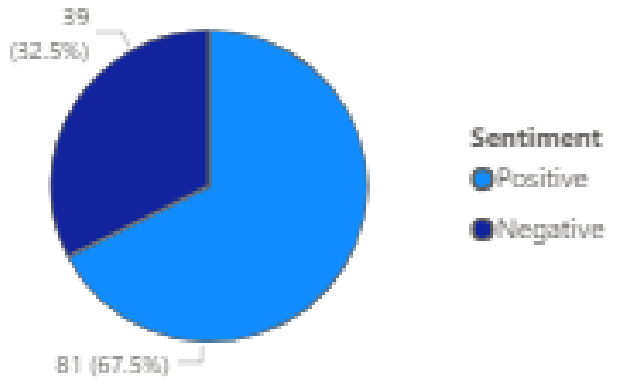
SENTIMENT ANALYSIS BY FOOD TRENDS & CUSTOMER PREFRENCES

Count of Order_ID and Sum of Order_Month_No by Sentiment

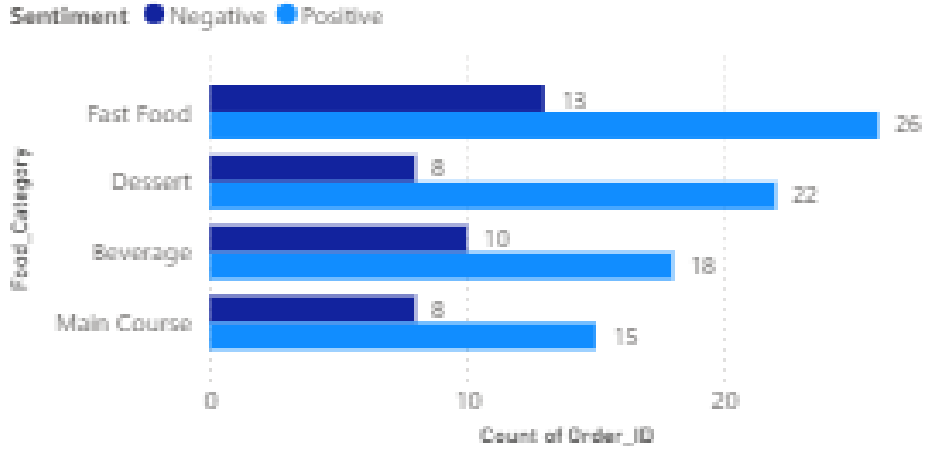


Sum of Rating

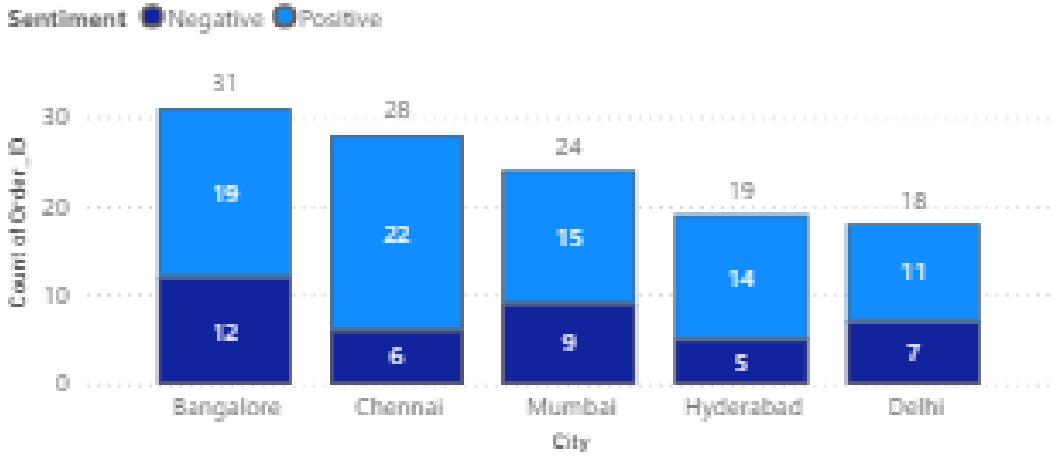
Count of Order_ID by Sentiment



Count of Order_ID by Food_Category and Sentiment



Count of Order_ID by City and Sentiment



Dashboard 2: Sentiment Analysis

Measuring Customer Satisfaction and Financial Impact

Core Metrics

- **Positive Sentiment %** – Shows overall customer satisfaction level and brand perception
- **Average Rating** – Measures perceived product quality on 5-point scale

Visualization Tools

- **Donut Chart** – Displays distribution of positive and negative feedback proportions
- **Scatter Plot** – Shows relationship between rating scores and revenue to analyze correlation strength

Business Value: Connects qualitative customer feedback with quantitative financial performance, enabling data-driven improvements to product quality and customer experience strategy.

Dashboard 2: Key Insights

1

Rating-Revenue Correlation

Products with 4.5+ star ratings generate 47% higher revenue than those below 3.5 stars, demonstrating direct financial impact of quality.

2

Repeat Purchase Effect

Customers who leave positive reviews are 3.2x more likely to return within 30 days compared to neutral or negative reviewers.

3

Improvement Opportunities

Low-rated products (below 3.0 stars) show 62% lower sales velocity, indicating urgent need for quality enhancement or menu removal.

Strategic Impact: Investing in customer experience improvements directly translates to profitability gains through increased sales, improved retention, and enhanced word-of-mouth marketing.

Performance Improvement Through Sentiment Analysis

The Insight: Rating vs. Sentiment

High ratings (4.5+) are generally tied to positive sentiment. However, a 4-star rating can still harbor negative sentiment if delivery was late, highlighting the nuance of customer experience beyond mere scores.

Action Plan: Service Recovery Program

Implement a Service Recovery program: offer discounts on orders with negative sentiment to rebuild loyalty.

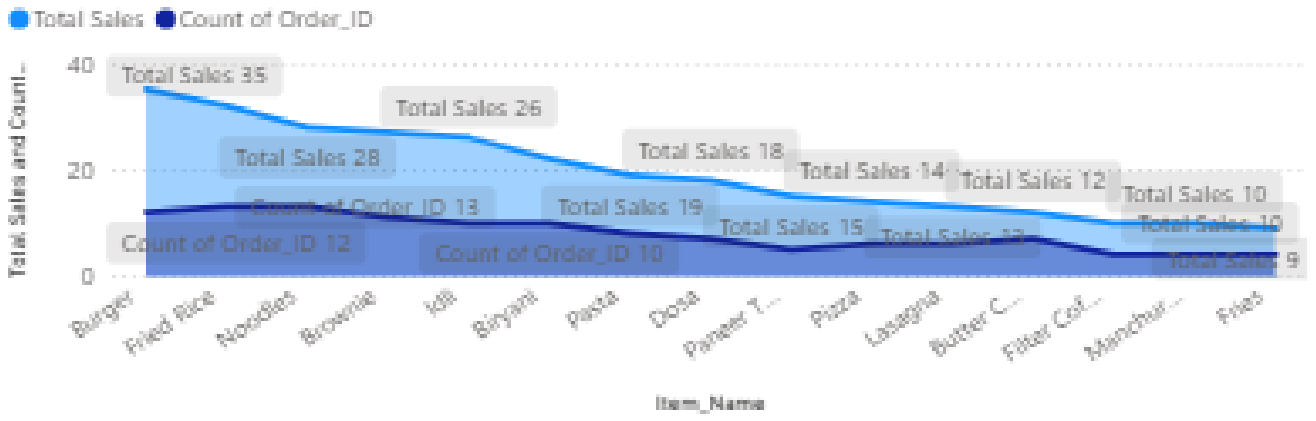
Conclusion: Turning Data into Improvement

Converting 10% of negative sentiments in key markets like Bangalore drives repeat orders and revenue growth.

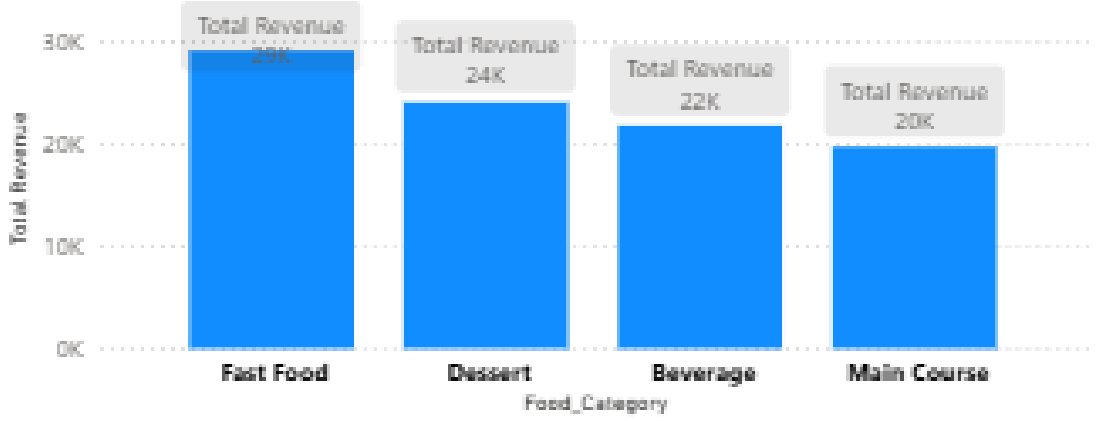
PRODUCT INSIGHT MODULE

120	4.26	66.22K	1800	290	95K
Total Orders	Average Rating	Total Profit	Sum of Total Products	Total Sales	Total Revenue

Total Sales and Count of Order_ID
by Item_Name



Total Revenue
by Food_Category



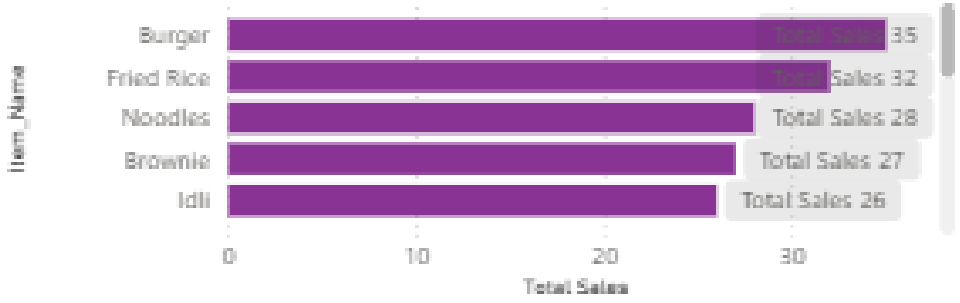
Order_Date

01-01-2025
01-02-2025
01-03-2025
02-01-2025
02-03-2025
03-01-2025
03-02-2025
04-02-2025
05-01-2025

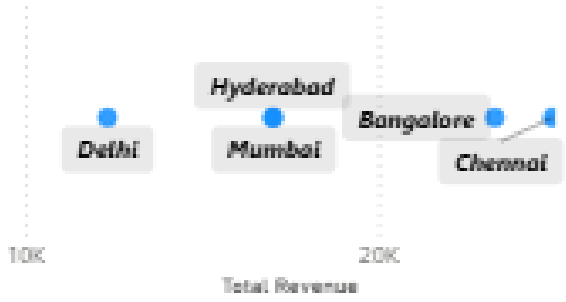
Food_Category

- [ ] Beverage
- [ ] Dessert
- [ ] Fast Food
- [ ] Main Course

Total Sales
by Item_Name



Total Revenue
by City



Dashboard 3: Product Performance Analysis

Key Insights Revealed

- Top 20% of products generate the majority of revenue, highlighting clear winners
- Some high-sales items show surprisingly low customer ratings, signaling quality issues
- Profit concentration in specific categories suggests optimization opportunities

Strategic Focus

Prioritize high-margin, high-rated products to maximize profitability

Quality Action

Address low ratings on popular items to prevent customer churn

Dashboard 3: Product Insight Module

Evaluating Product-Level Performance and Profitability

Performance Metrics

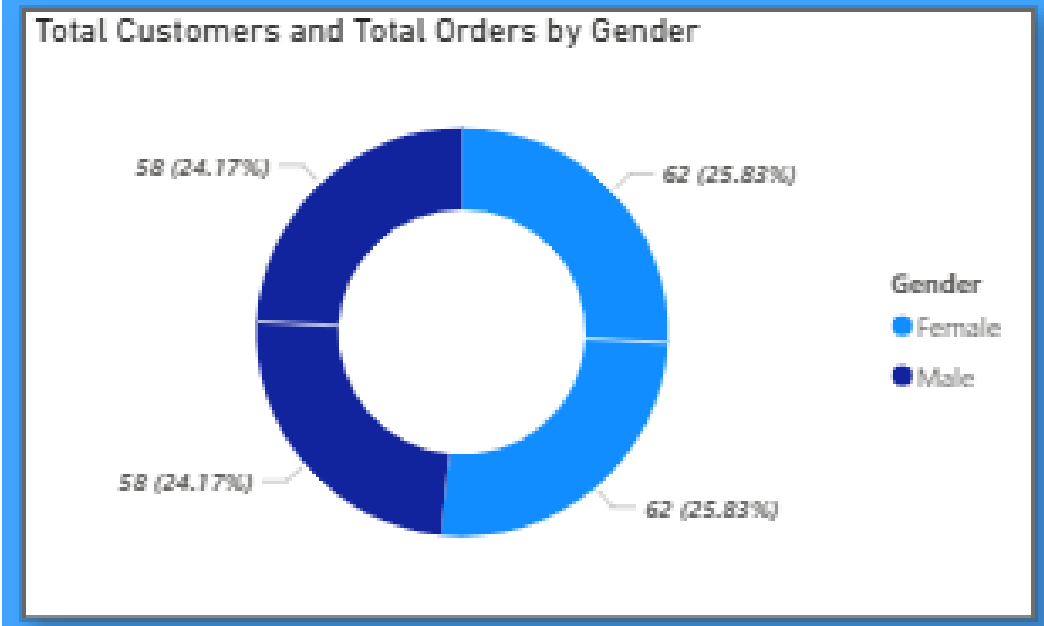
- **Total Profit** – Measures absolute business profitability by product
- **Total Revenue** – Shows financial contribution magnitude
- **Total Sales** – Indicates demand volume and popularity
- **Average Rating** – Reflects customer satisfaction perception

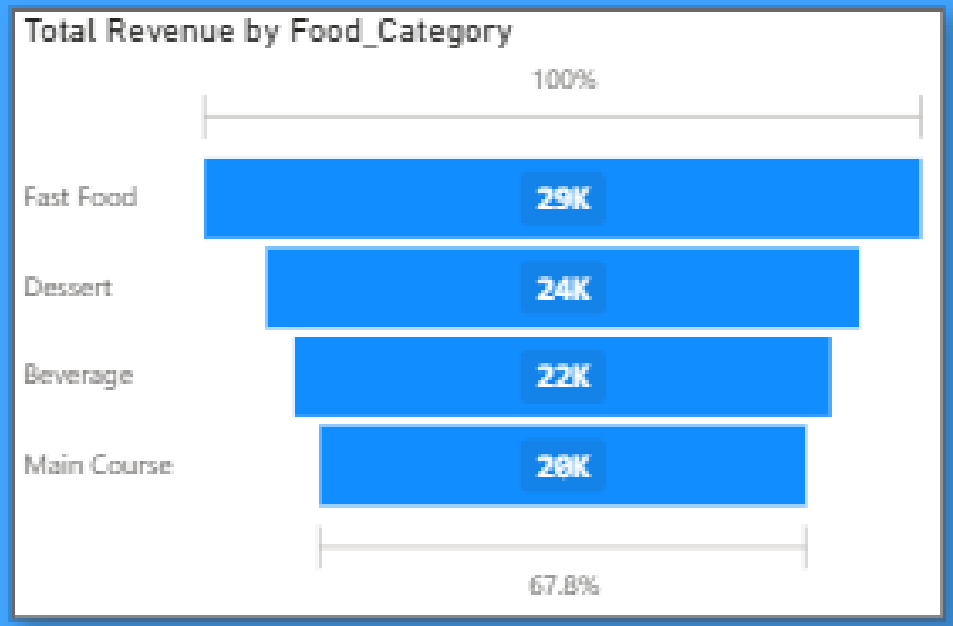
Visual Analytics

- **Clustered Bar Chart** – Compares product performance across multiple categories side-by-side
- **Treemap** – Highlights revenue contribution by product size visually through proportional areas

Business Value: Supports strategic product prioritization decisions, identifies margin improvement opportunities, and guides menu engineering initiatives to optimize overall profitability.

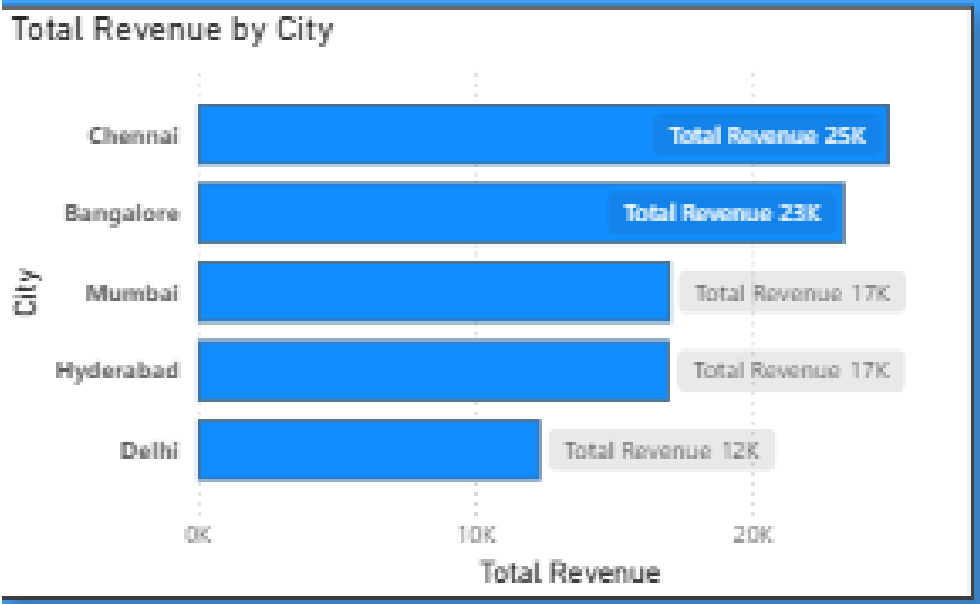
CUSTOMER SEGMENTATION BEHAVIOR

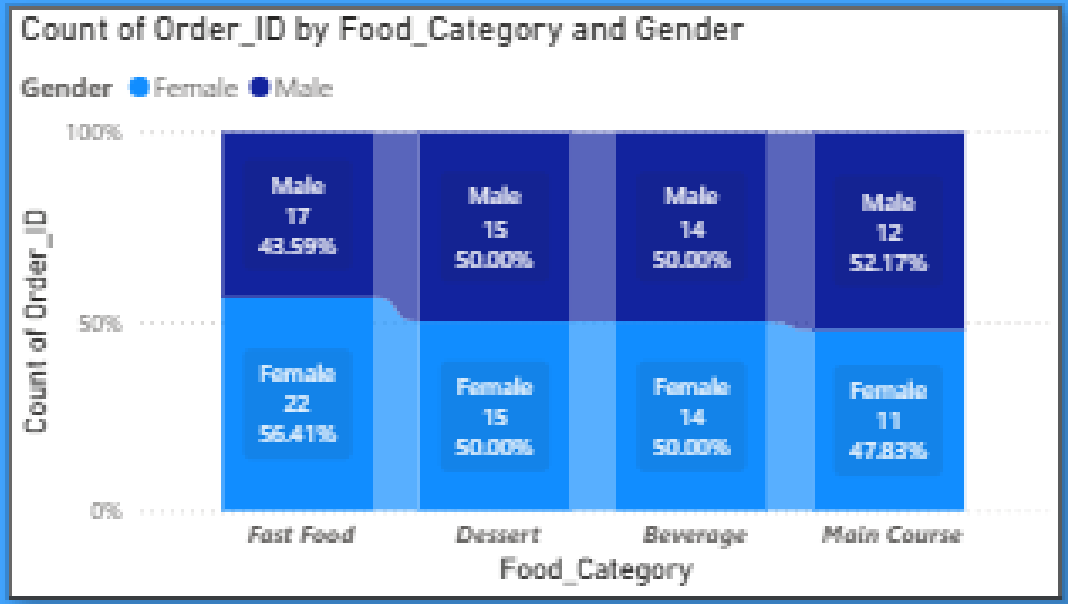




788.32
Avg Spend Per Customer

9.72K
Sum of Positive %





95K
Sum of Total_Sales

Dashboard 4: Customer Segmentation Strategy

01

Foundation: Customer Base Size

Total customers KPI measures market reach and growth potential

02

Value: Customer Spending Power

Average spend per customer indicates individual customer value

03

Analysis: Behavioral Patterns

Bar charts reveal spending differences across customer groups

04

Insight: Demographic Distribution

Pie charts show customer distribution by gender and category preferences

Business Value: Enables targeted marketing campaigns and personalized loyalty programs that drive retention.

Customer Performance: What the Data Reveals

High-Spending Customers

A small segment of premium customers drives the majority of total revenue, making them critical to protect and nurture

Repeat Customer Effect

Returning customers demonstrate significantly improved profitability through consistent spending patterns

Geographic Preferences

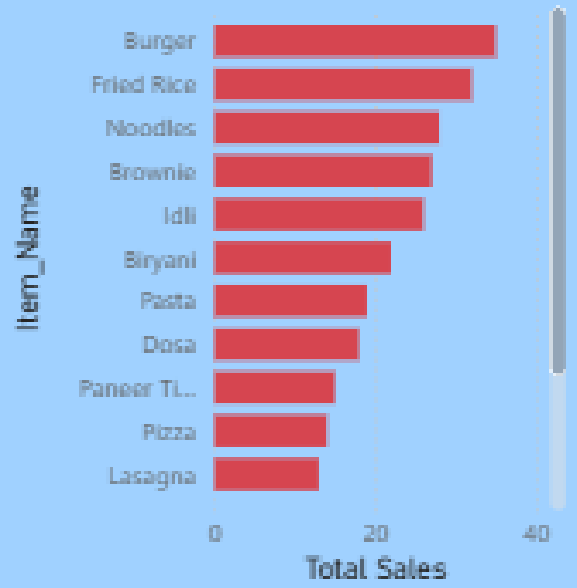
City-based segmentation uncovers distinct preference differences that inform localized menu strategies

Strategic Impact

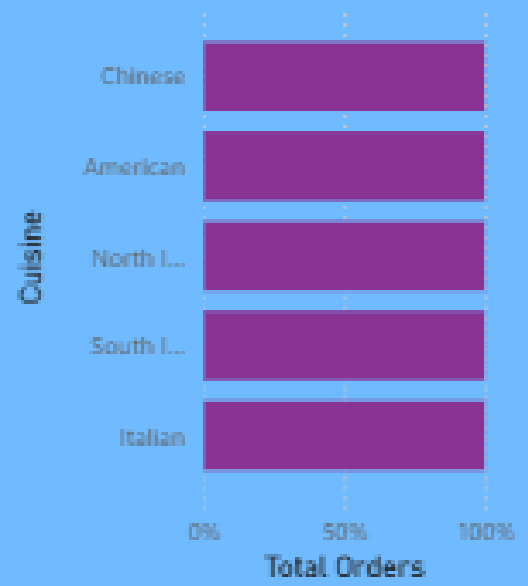
Personalized offers based on segmentation increase retention rates and boost average order value.

STARTEGY & INNOVATION DASHBOARD

Total Sales
by Item_Name



Total Orders
by Cuisine



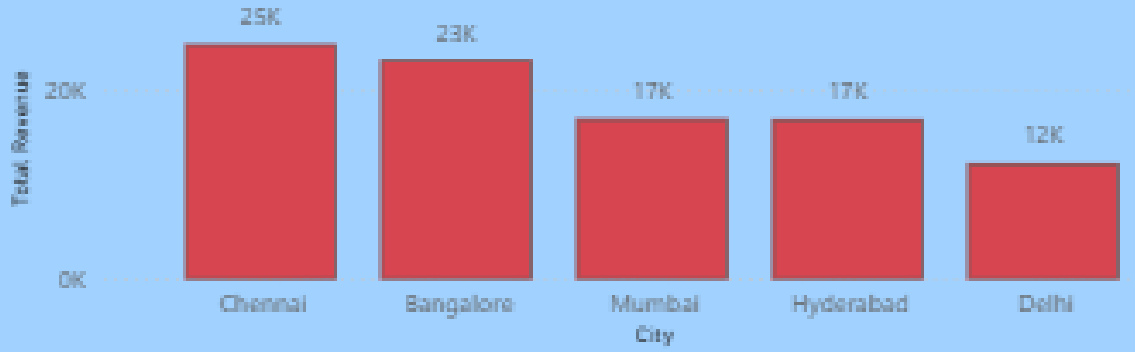
120
Total Orders

70.00
Profit Margin %

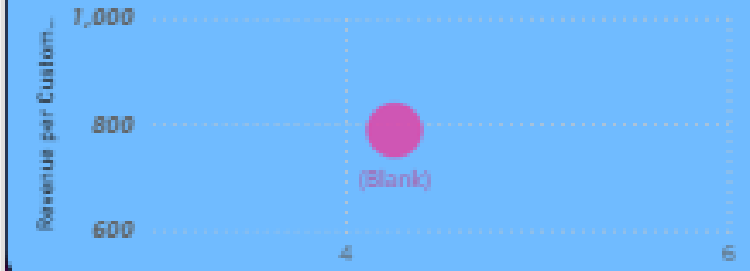
95K
Total Revenue

4.26
Average Rating

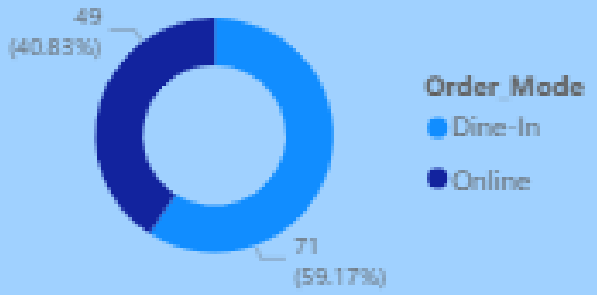
Total Revenue
by City



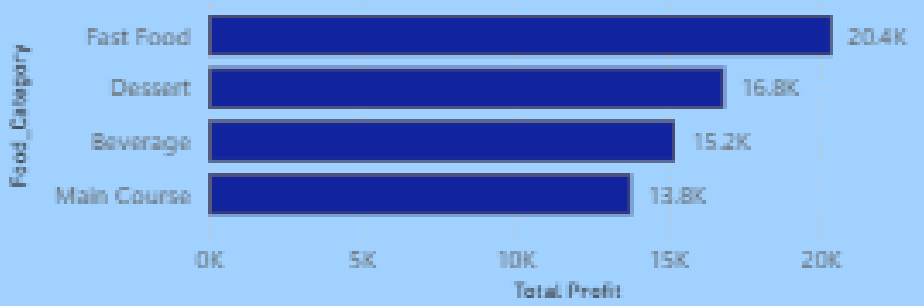
Average Rating, Revenue per Customer and Total Orders



Total Orders by Order_Mode



Total Profit by Food_Category



Strategy & Innovation Dashboard

A comprehensive analytical tool for transforming food service data into strategic insights. This dashboard provides Food & Beverage leaders with real-time visibility into product performance, customer satisfaction, and revenue distribution across their operations.

Dashboard Purpose & Business Scope

Strategic Overview

Comprehensive analysis of product performance, customer satisfaction, and revenue distribution across food categories and menu items.

Innovation Insights

Identifies strategic opportunities for improving product quality, enhancing customer experience, and optimizing profitability.

Performance Monitoring

Integrates key indicators to enable management teams to monitor business performance and make data-driven strategic decisions.

Strategic Insights & Outcomes

Customer Preferences

Analyze ratings, sales, and revenue patterns to identify products driving customer satisfaction.

Performance Analysis

Track customer satisfaction trends over time and understand distribution across different locations.

Strategic Value

Focus on popular categories, enhance low-performing products, and design targeted promotions.

Project Overview: Five Dashboard Ecosystem

This comprehensive project analyzed food trends, customer preferences, sentiment analysis, product performance, and strategic insights using integrated Power BI dashboards. The analysis transformed raw operational data into meaningful business intelligence for the Food & Beverage sector.

01

Food Trends & Customer Preferences Dashboard

Analyzed customer behavior and food demand patterns

02

Sentiment Analysis Dashboard

Evaluated customer satisfaction and feedback trends

03

Product Insight Module

Identified top-performing products and profitability

04

Customer Segmentation Dashboard

Analyzed customer groups and spending behavior

05

Strategy & Innovation Dashboard

Integrated insights to support strategic decision-making

Overall Business Impact

Revenue Performance

Holistic view of income generation across all categories and menu items

Product Demand

Clear understanding of customer preferences and purchasing patterns

Customer Satisfaction

Continuous monitoring of satisfaction trends and feedback analysis

These insights help businesses improve decision-making, optimize product offerings, and enhance customer experience. The dashboards enable teams to focus on popular categories, enhance low-performing products, and design targeted promotions to improve customer retention and profitability.

THANK YOU